

Before the Levy: Announcement Pass-Through and Carbon-Price Calibration in Denmark’s Livestock Tax

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Abstract

Denmark announced in June 2024 that it would introduce a greenhouse-gas tax on livestock emissions, making the agreement one of the first major attempts to price agricultural emissions directly. This draft studies whether that announcement affected supermarket prices before the tax was implemented. Using Danish grocery price histories and a difference-in-differences (DiD) design with product-store and period fixed effects, I find that beef prices increased by 5.8 log points, or about 6.0 percent, after the announcement. Beef is the only treated meat category that both shows a positive price response and passes the individual pre-trend screen in the event-study specification. A synthetic difference-in-differences robustness check gives a larger but much less precise estimate because the method requires a complete commodity-store panel. I then use the estimated beef effect in a back-of-the-envelope carbon-price calibration. After growing the Nordhaus (2017) social cost of carbon from 2015 to 2030 and converting to 2024 dollars, the DiD announcement effect is equivalent to about USD 23.2/tCO₂e. The implied log-price reference needed to close the remaining carbon-price gap lies inside the DiD confidence interval, so the estimates do not reject exact compensation at the 95 percent level. Point estimates nevertheless illustrate how the framework can fine-tune policy: under the DiD ATT, a statutory gross tax of about USD 46.8/tCO₂e, or DKK 322/tCO₂e, would exactly match the inflation-adjusted Nordhaus benchmark after accounting for announcement pass-through, while the latest event-study point estimate would imply lowering the beef tax if anticipatory effects keep increasing.

1 Introduction

Agricultural greenhouse-gas emissions remain among the hardest emissions to price. Livestock emissions are biologically and politically salient, consumer prices are highly visible, and pass-through may begin before a formal tax is collected if firms and consumers update expectations when policy is announced. This paper asks whether Denmark’s livestock carbon-tax announcement affected supermarket beef prices before implementation, and whether the resulting price change is large enough to matter for the calibration of the eventual tax.

On June 24, 2024, the Danish government and the parties to the Green Tripartite announced an agreement for a “green Denmark” (Danish Government, 2024). Public coverage emphasized that

Denmark would become the first country to impose a carbon tax on livestock emissions. Institutionally, the policy is not a retail meat excise tax. It is an upstream agricultural greenhouse-gas tax, with implementation scheduled for later years and with deductions and transition mechanisms. Public attention focused especially on cattle and pigs, while the broader agreement also covers land-use and agricultural restructuring.

The main empirical result is that beef prices increased after the announcement. In the preferred difference-in-differences specification, the estimated average treatment effect for beef is 0.0583 log points, with a standard error of 0.0157. Interpreted as a percentage price change, this is approximately 6.0 percent. The event-study evidence for beef shows no statistically significant individual pre-period coefficient at the 95 percent level, whereas the other treated meat categories fail the same pre-trend screen. For that reason, the main analysis focuses on beef.¹

The second contribution is a simple carbon-price calibration. Nordhaus (2017) reports a 2015 social cost of carbon of USD 31.2/tCO₂ in 2010 dollars and a real growth rate of 3 percent per year through 2050. Growing that value to 2030 and converting it to 2024 dollars gives a benchmark of about USD 69.9/tCO₂e. Denmark’s announced gross tax of DKK 300/tCO₂e is about USD 43.5/tCO₂e at the 2024 average exchange rate, leaving an internally consistent gap of about USD 26.4/tCO₂e. The DiD beef announcement effect is equivalent to about USD 23.2/tCO₂e. Since the gap-implied log effect lies inside the estimated confidence interval, these data do not reject the hypothesis that announcement pass-through already compensates the gap at the 95 percent level. Point estimates are still policy-relevant: if policymakers wished the future statutory tax plus announcement-induced pass-through to exactly match the inflation-adjusted Nordhaus benchmark, the DiD ATT implies a statutory gross tax of about USD 46.8/tCO₂e, or DKK 322/tCO₂e. If anticipatory beef price effects continue to increase, as the event-study path suggests, the beef-specific tax may instead need to be lowered.

The broader interpretation is that policy announcements may themselves be economically powerful instruments. The logic is close to the celebrated “whatever it takes” moment in monetary policy: a credible public commitment can shift expectations before the full policy instrument is deployed (Draghi, 2012). That analogy is not exact, but it highlights the same mechanism. If firms and consumers react to credible future policy, some intended price and expectation effects may occur before the statutory instrument takes effect. That does not eliminate the need for policy; rather, it gives policymakers a reason to measure anticipatory pass-through and recalibrate future policy in light of observed market adjustment.

2 Data and Panel Construction

The analysis uses Danish grocery price histories from the dagligepriiser.dk data ecosystem. The pipeline downloads product-level price histories, expands each price-history entry into a product-

¹The pipeline also estimates effects for pork, lamb/sheep/goat, mixed livestock products, dairy cattle, and aggregate treated products. These estimates are useful diagnostics and appendix material, but the pre-period evidence does not support treating them as the main causal estimates.

date observation, classifies products by commodity, and normalizes prices to DKK per kilogram or liter when source quantity information permits. Rows with failed price normalization are excluded from the main estimating sample. The main sample is food-only and excludes unknown commodity classifications.

The baseline panel is monthly. The event date is June 24, 2024; in the monthly panel, June 2024 is treated as the event period and excluded from the pre/post comparison. The estimation unit is a product-store pair. Units are retained if they have at least one pre-event and one post-event observation. This inclusion rule preserves the large grocery cross-section while avoiding the extreme attrition that would result from requiring every product-store pair to appear in every month.

Treatment follows the institutional design of the policy. Beef, pork, lamb/sheep/goat, mixed livestock products, and dairy cattle are coded as livestock-exposed treated groups in the main pipeline. Non-meat foods, fish, and seafood serve as controls. The paper focuses on beef because it is both a prominent target of the policy announcement and the only meat category with a positive post-announcement estimate and acceptable pre-period event-study diagnostics.

3 Identification Strategy

The preferred specification is a two-way fixed-effects difference-in-differences model:

$$\log(p_{it}) = \alpha_i + \lambda_t + \beta_g (\mathbf{1}\{g_i = \text{beef}\} \times \text{Post}_t) + \varepsilon_{it}, \quad (1)$$

where p_{it} is normalized price for product-store unit i in month t , α_i are unit fixed effects, and λ_t are month fixed effects. For commodity-specific estimates, the sample contains the focal treated commodity and untreated food controls only; other treated livestock commodities are excluded from that commodity's regression. Standard errors are clustered by product-store unit. The identifying assumption is that, absent the announcement, beef prices would have followed the same time path as the control food products after accounting for product-store and period fixed effects.

I assess this assumption with an event-study design:

$$\log(p_{it}) = \alpha_i + \lambda_t + \sum_{\tau \neq -1} \delta_\tau (\mathbf{1}\{g_i = \text{beef}\} \times \mathbf{1}\{\text{RelativeTime}_t = \tau\}) + \varepsilon_{it}. \quad (2)$$

The omitted period is $\tau = -1$, which is plotted as a zero estimate with no confidence interval. The beef pre-period coefficients are small relative to the post-period movement and pass the individual 95 percent pre-trend screen.

As a robustness check, I also estimate synthetic difference-in-differences ([Arkhangelsky et al., 2021](#)). The SDiD estimator constructs unit weights to match the treated pre-period path and time weights to adjust pre-period imbalance. This estimator is applied to complete commodity-store units rather than the full product-store panel, because the weighting step requires a rectangular panel. The resulting beef estimate is larger than the DiD estimate, but its confidence interval is wide, reflecting the reduction to 11 complete commodity-store units.

4 Results

Table 1 summarizes the main DiD estimates across treated categories. Beef is the central estimate for the paper: the coefficient is 0.0583 with a standard error of 0.0157, statistically significant at the 1 percent level. The implied price increase is $\exp(0.0583) - 1 = 6.0$ percent.

Table 1: DiD announcement effects

	All treated	Beef	Dairy cattle	Lamb, sheep, goat	Mixed livestock	Pork
Treatment $\times$ Post	0.0138*** (0.0045)	0.0583*** (0.0157)	0.0344*** (0.0046)	-0.0845*** (0.0323)	-0.0001 (0.0406)	-0.0358*** (0.0067)
Observations	287,480	207,369	252,903	203,630	206,215	220,563
Units	22,780	16,912	20,067	16,525	16,706	17,850
Periods	32	32	32	32	32	32
Pre-treatment avg.	257.48	158.54	279.50	861.93	135.86	186.35
R^2	0.000	0.000	0.000	0.000	0.000	0.000

Notes: Outcome is log normalized price in DKK per kilogram or liter. The all-treated column compares all treated livestock-exposed commodities with untreated food controls. Each commodity column is estimated separately against untreated food controls only. All columns include unit and period fixed effects. Standard errors, in parentheses, are clustered by unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 1 shows the beef event study. The pre-period estimates do not display a clear differential trend and the omitted period $\tau = -1$ is shown at zero. The post-period path rises over time, with especially large effects in the latest months.

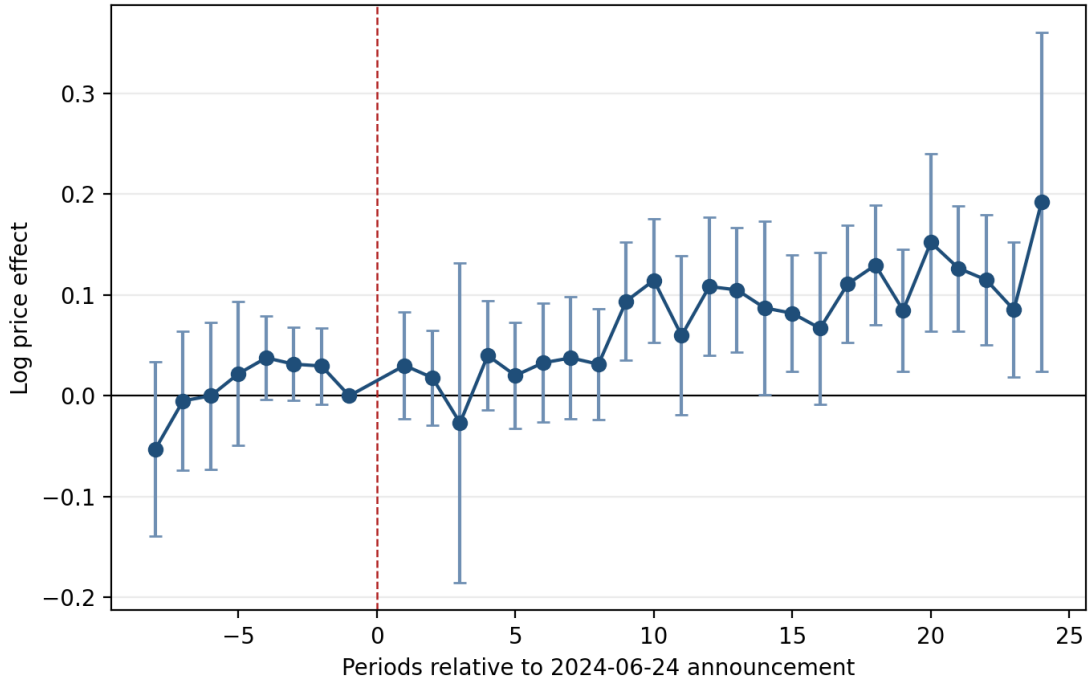


Figure 1: Beef event-study estimates

The SDiD robustness check in Table 2 gives a beef estimate of 0.1828 with a standard error of 0.0926. Although this estimate is larger than the DiD estimate, it is much less precise. The SDiD

confidence interval overlaps the DiD confidence interval, so the data do not allow us to rule out equality of the two estimates at conventional levels. The imprecision is substantively important: the SDiD design keeps only complete commodity-store units, reducing the beef SDiD sample to 352 observations and 11 units.

Table 2: Synthetic DiD announcement effects

	All treated	Beef	Dairy cattle	Mixed livestock	Pork
Treatment $\times$ Post	0.0502 (0.0348)	0.1828** (0.0926)	0.0197 (0.0351)	-0.0100 (0.0184)	-0.0118 (0.0241)
Observations	512	352	352	320	352
Units	16	11	11	10	11
Periods	32	32	32	32	32
Pre-treatment avg.	123.21	153.52	127.78	95.52	102.17
R^2					

Notes: Synthetic DiD uses complete commodity-store units. Unit weights match the treated pre-period path; time weights match post-period donor averages. Standard errors, in parentheses, use a nonparametric bootstrap over complete treated and control units. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Carbon-Price Calibration

The calibration asks what statutory tax would exactly match the Nordhaus benchmark once the observed announcement-induced price increase is taken into account. Let $\hat{\tau}$ denote an estimated beef log price effect and $\bar{p}_{pre}$ the average pre-announcement beef price in DKK/kg. The beef carbon intensity is set to 59.6 kg CO₂e/kg product, using the OECD agri-food carbon-footprint report’s presentation of the [Poore and Nemecek \(2018\)](#) beef-herd benchmark ([OECD, 2025](#)). The announcement-equivalent carbon price is:

$$\widehat{SCC}_{announce} = \frac{[\bar{p}_{pre} (\exp(\hat{\tau}) - 1) / 6.8953] \times 1000}{59.6}. \quad (3)$$

The statutory tax that exactly reaches the benchmark is:

$$Tax^* = SCC_{Nordhaus, 2030}^{2024USD} - \widehat{SCC}_{announce}. \quad (4)$$

[Nordhaus \(2017\)](#) reports USD 31.2/tCO₂ for 2015 in 2010 dollars and a real growth rate of 3 percent per year. Therefore:

$$SCC_{Nordhaus, 2030}^{2024USD} = 31.2(1.03)^{15} \frac{313.689}{218.056} = 69.9. \quad (5)$$

The announced Danish gross tax is $300/6.8953 = 43.5$ USD/tCO₂e, so the benchmark gap is 26.4 USD/tCO₂e.

Figure 2 compares the DiD and SDiD beef estimates to the log price increase needed to close that gap. The reference value lies inside both confidence intervals. Therefore, the data do not reject the hypothesis that the anticipatory beef price effect already compensates the tax gap at

the 95 percent level. This is the main inferential takeaway. The reverse-engineered point estimate should be read as a policy-calibration exercise rather than as a statistically sharp recommendation: using the preferred DiD estimate, the announcement effect is equivalent to 23.2 USD/tCO₂e, so the tax that exactly matches the inflation-adjusted Nordhaus benchmark is 46.8 USD/tCO₂e, or DKK 322/tCO₂e. This is slightly above, not below, the announced DKK 300/tCO₂e gross tax.

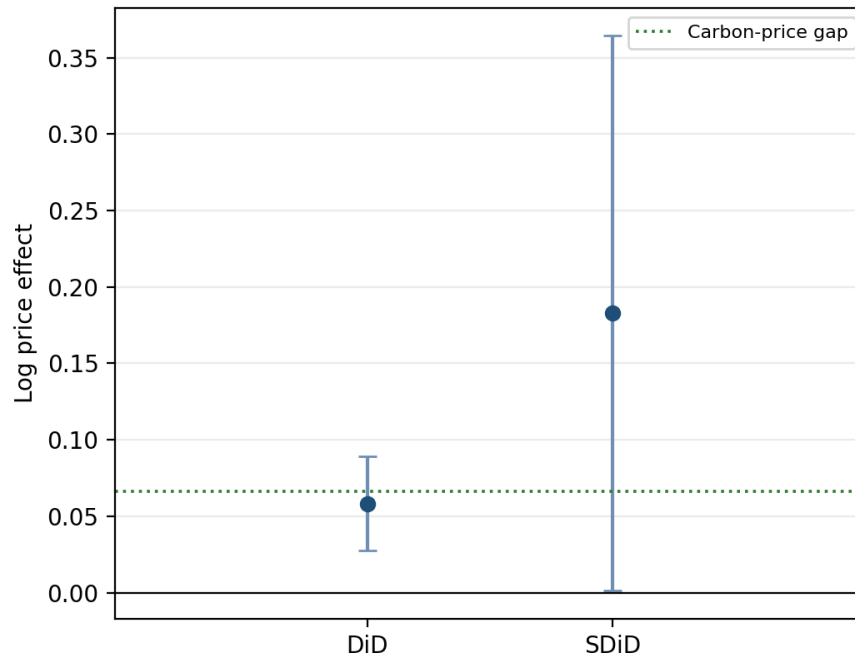


Figure 2: Beef ATT estimates and carbon-price-gap reference

The latest event-study estimate gives a different policy implication as a point-estimate exercise. At relative period 24, the beef estimate is 0.1924. This corresponds to 81.9 USD/tCO₂e, exceeding the inflation-adjusted Nordhaus benchmark. The literal reverse-engineered tax is therefore negative, −11.9 USD/tCO₂e, or −82 DKK/tCO₂e. With a nonnegative tax constraint, the implied statutory tax is zero. The confidence interval for this latest-period estimate also contains the gap reference, so the policy conclusion should remain cautious. Still, if beef pass-through continues to rise, then the product-specific tax needed to align beef prices with the benchmark would likely be lower than the announced gross rate.

6 Discussion and Conclusion

The evidence suggests that Denmark’s livestock-tax announcement affected beef prices before the tax was implemented. This is consistent with anticipatory pass-through: firms and consumers may update expectations, contracts, and pricing strategies when a future carbon price becomes credible. Such anticipation complicates policy evaluation because the announcement and implementation stages are not cleanly separable.

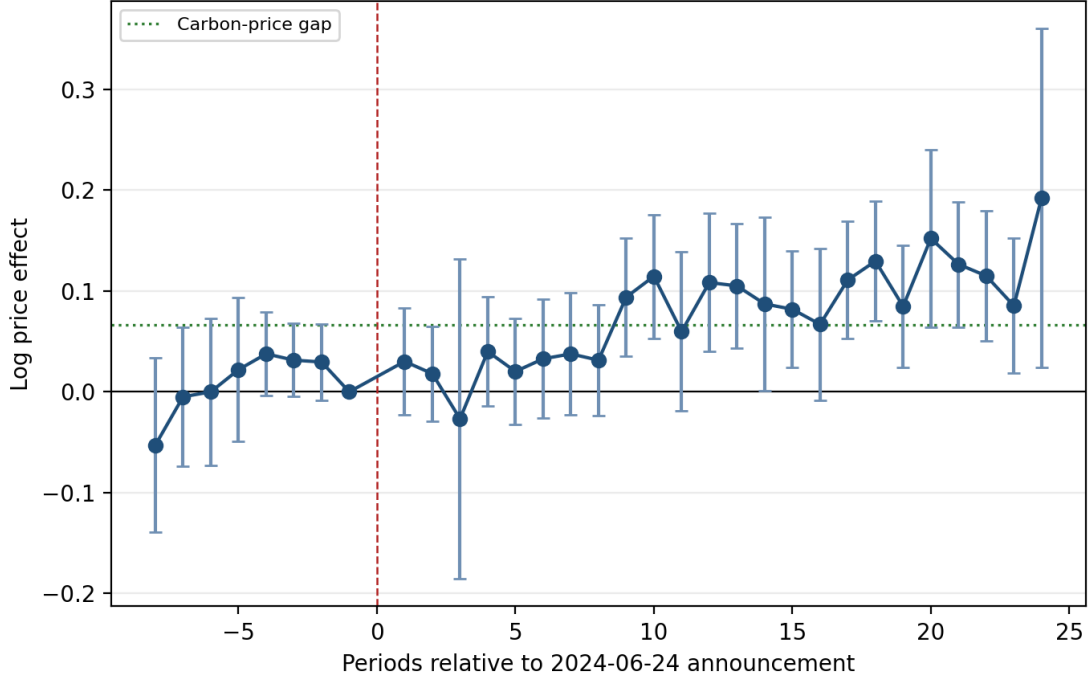


Figure 3: Beef event-study estimates and carbon-price-gap reference

For policy design, the central lesson is not that announcements substitute for taxes. Rather, announcements can generate measurable market adjustment that should be incorporated into later calibration. Under the preferred DiD ATT, the announcement-induced beef price effect covers a large share of the gap between the announced Danish tax and the inflation-adjusted Nordhaus benchmark. Because the gap lies inside the ATT confidence interval, however, we cannot rule out exact compensation at the 95 percent level. Under the latest event-study point estimate, the announcement effect alone exceeds the benchmark, implying a nonnegative tax floor of zero. These two scenarios should be distinguished carefully: the ATT is the more stable summary for policy calibration, while the latest-period estimate is useful as a warning that anticipatory effects may grow over time.

Several limitations remain. Product classification is deterministic and should receive manual review for mixed meat products. The analysis studies supermarket prices rather than farm-level emissions or producer incidence. Finally, the carbon-price calibration is an accounting exercise, not a structural model of pass-through. Even so, it illustrates a useful policy principle: credible announcements can be policy instruments in their own right, and observing the market response to an announcement can help policymakers adjust the eventual statutory instrument before implementation. This contribution is especially useful for commodity-specific policy design. In the current evidence, beef is the only product for which causal interpretation is supported by both an observed effect and acceptable pre-trend diagnostics; for other livestock commodities, a uniform tax may not generate the same retail-price pass-through.

References

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A Additional Results

This appendix reports diagnostic outputs for the broader treated sample. These results are useful for transparency and robustness, but the main causal and calibration discussion remains focused on beef because the other treated commodities do not jointly show the required pre-period diagnostics and post-announcement pattern.

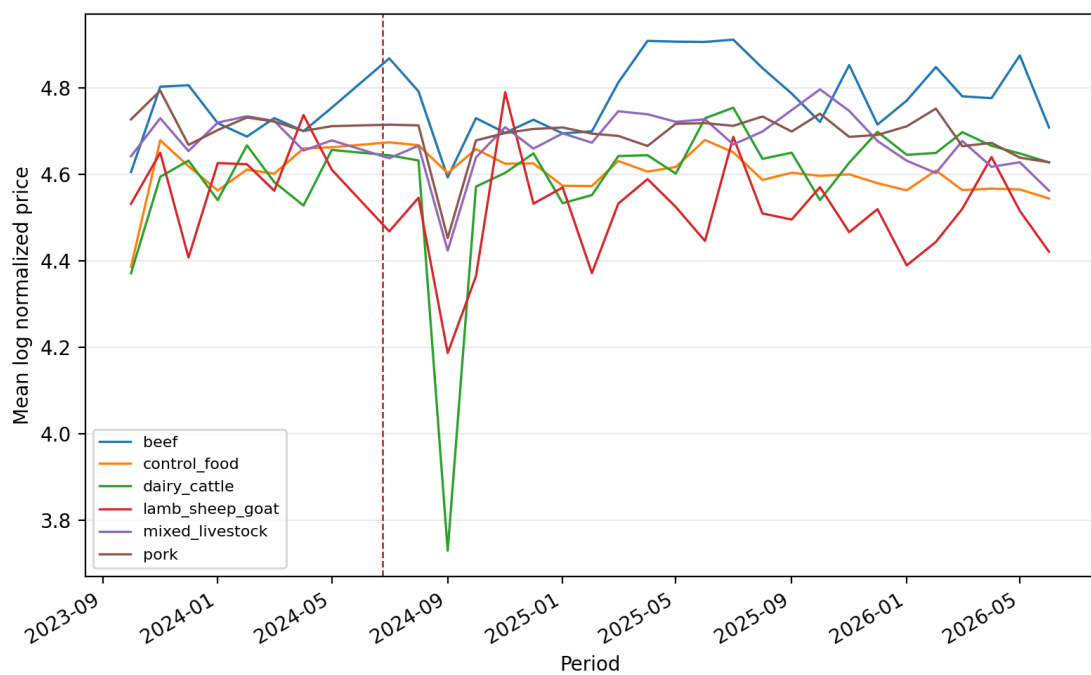


Figure 4: Aggregate treated and control price trends

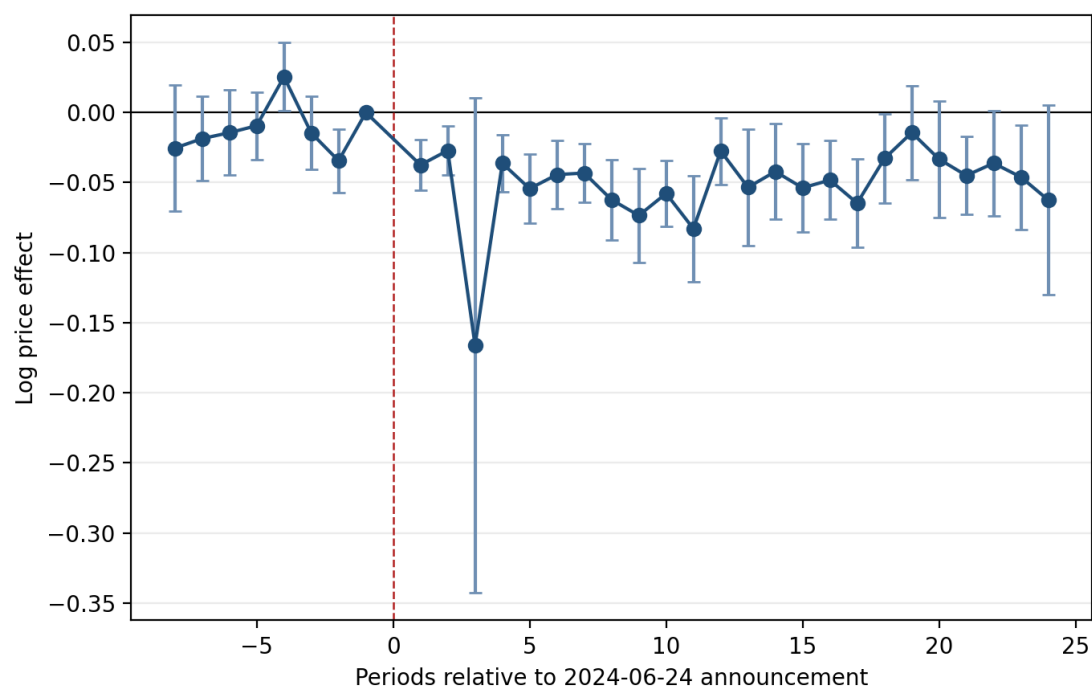


Figure 5: Pork event-study estimates

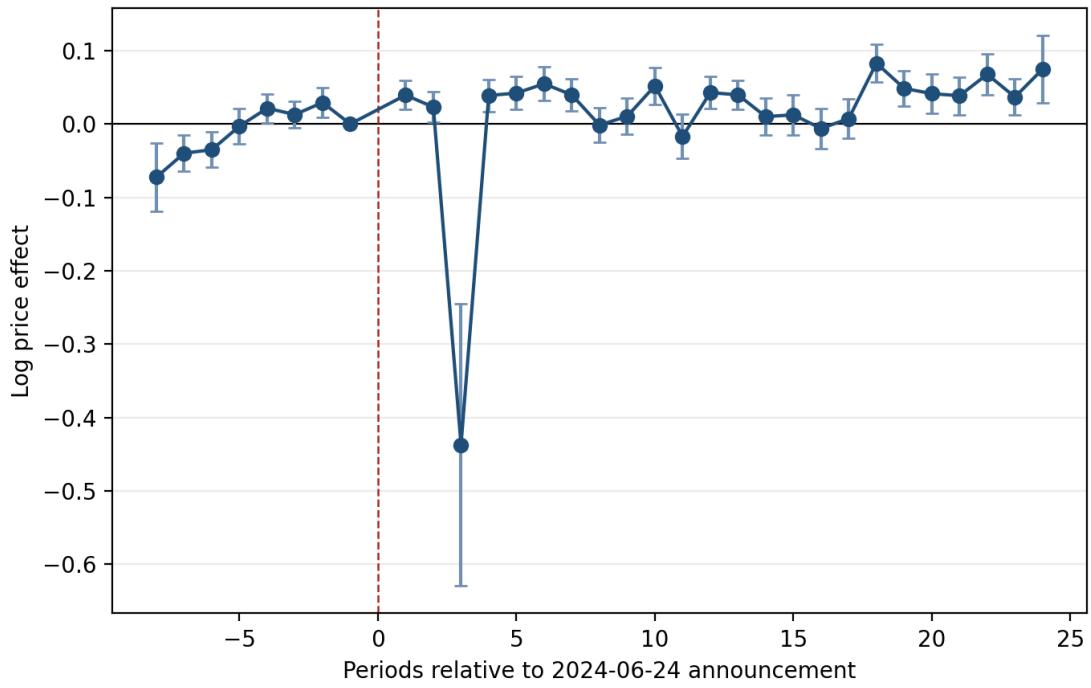


Figure 6: Dairy cattle event-study estimates

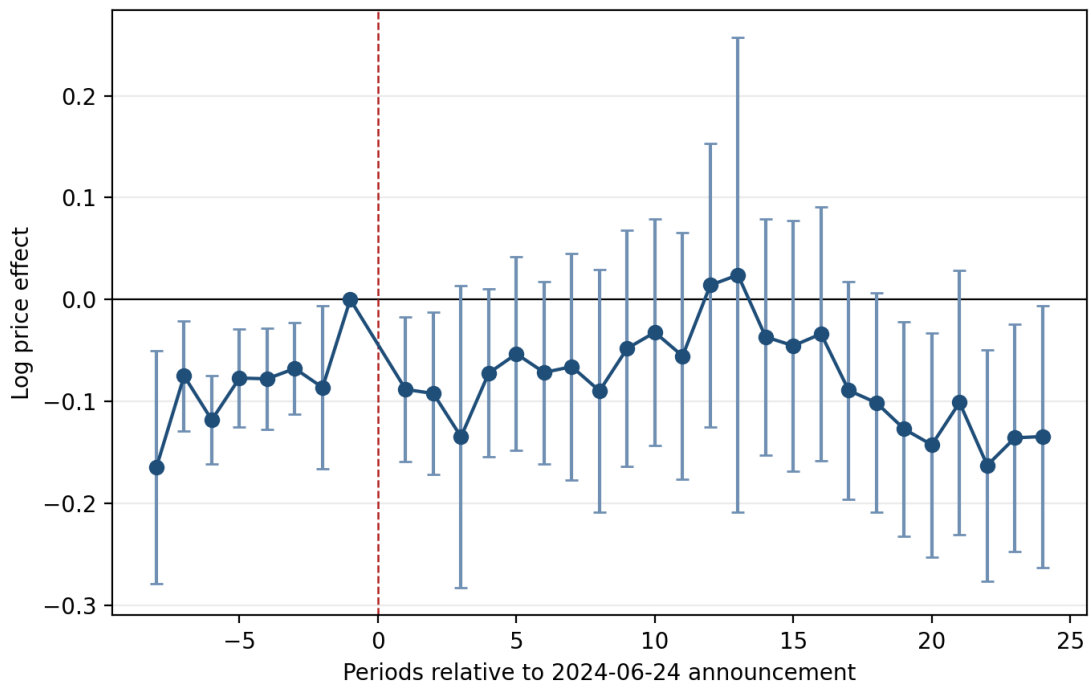


Figure 7: Mixed livestock event-study estimates

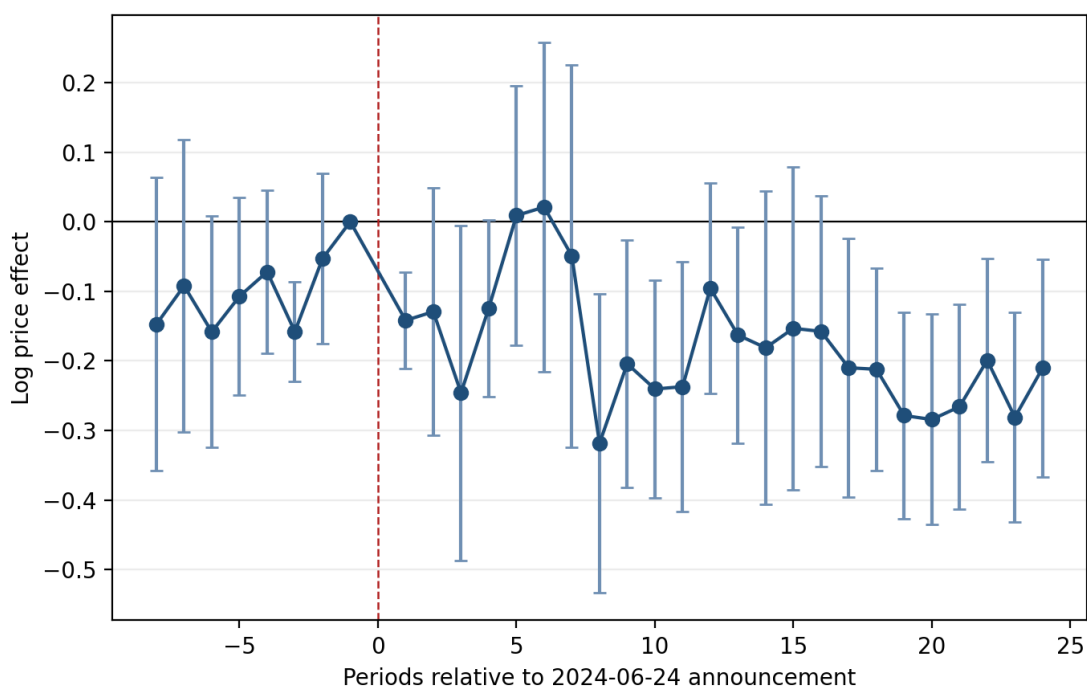


Figure 8: Lamb, sheep, and goat event-study estimates

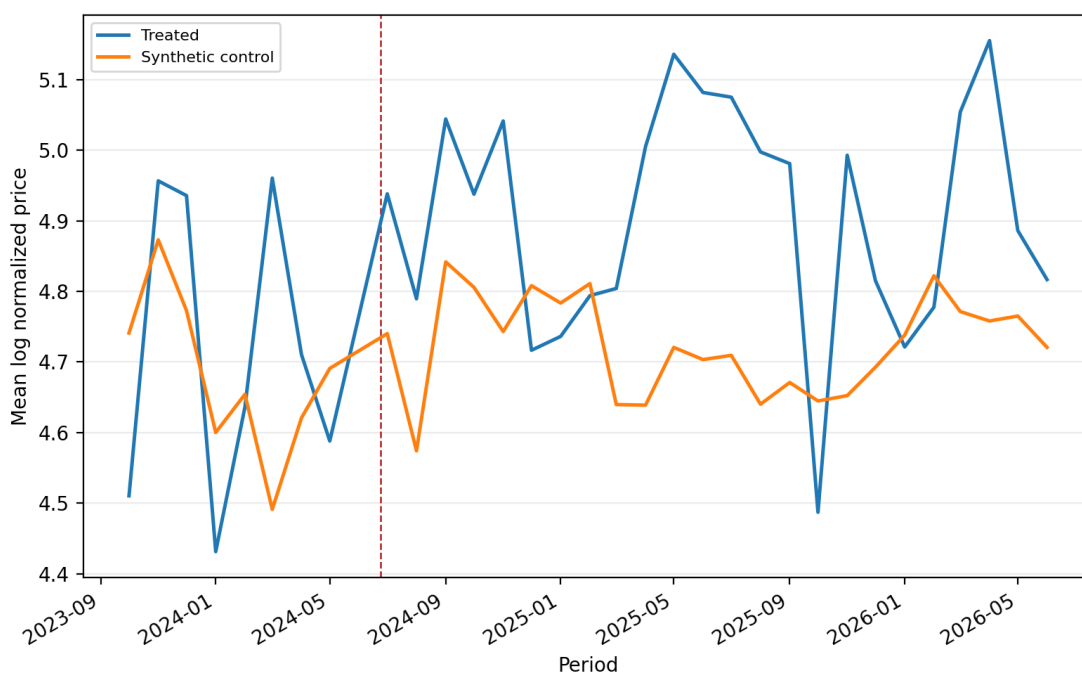


Figure 9: Synthetic DiD beef trends